

ELE214 - Electronics Laboratory I Preliminary Work 4

1 Questions

1. Explain the use of each circuit element in the figure briefly.

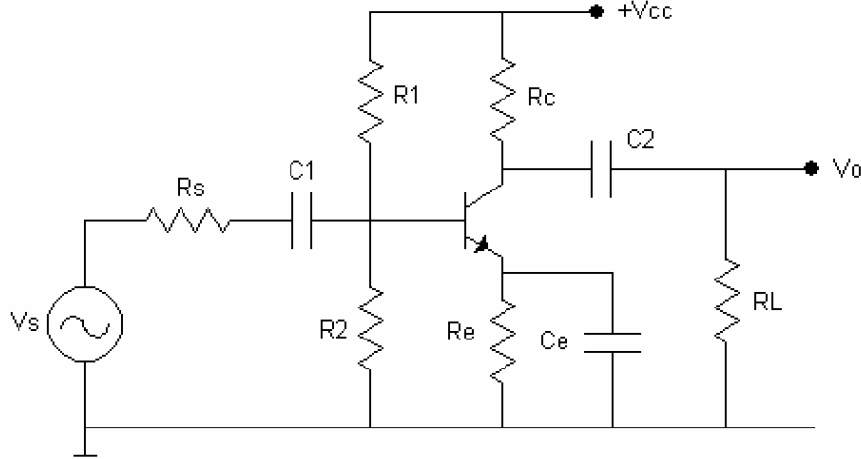


Figure 1

2. Design an amplifier to produce a maximum swing with the following parameters: AC voltage gain: 50, $V_{CC} = 15\text{ V}$, $h_{fe} = 100$, $V_{ce, \text{sat}} = 0.2\text{ V}$.

Experimental Work (Simulation of 1-4)

1. Set up the circuit in *Figure 1* with the following component values:

$$R_s = 22\text{ k}\Omega \quad R_1 = 56\text{ k}\Omega \quad R_2 = 12\text{ k}\Omega$$

$$R_3 = R_c = 3.3\text{ k}\Omega \quad R_e = 1\text{ k}\Omega \quad R_L = 2.2\text{ k}\Omega$$

$$C_1 = C_2 = C_3 = 10\text{ }\mu\text{F} \quad V_{CC} = 15\text{ V} \quad V_s = 1\text{ kHz sinus}$$

2. Measure the $I_{C,Q}$ and $V_{CE,Q}$ of the circuit.
3. Taking $V_s = 3V_{pp}$, obtain the transfer characteristics (V_o vs V_s) on the oscilloscope.
4. Taking $V_s = 0.5V_{pp}$, measure V_o with and without R_L .

2 Answers

1. V_s : The AC input signal to be amplified.

R_s : Source resistor.

R_L : Load resistor.

R_c : Collector resistor.

R_e : Emitter-degeneration resistor stabilizing the circuit.

R_1, R_2 : Resistors that help the transistor work in the active region, or simply biasing.

C_1, C_2 : Filter capacitors allowing AC input to pass and block DC input.

C_e : Emitter bypass capacitor increasing gain.

2. To achieve maximum symmetrical swing, the collector-emitter voltage V_{CEQ} is centered:

$$V_{CEQ} = \frac{V_{CC} + V_{ce,sat}}{2} = \frac{15\text{V} + 0.2\text{ V}}{2} = 7.6\text{ V}$$

Assuming $I_{CQ} = 1\text{ mA}$, the thermal resistance r'_e is:

$$r'_e = \frac{26\text{ mV}}{1\text{ mA}} = 26\ \Omega$$

The gain for the swamped amplifier is defined by:

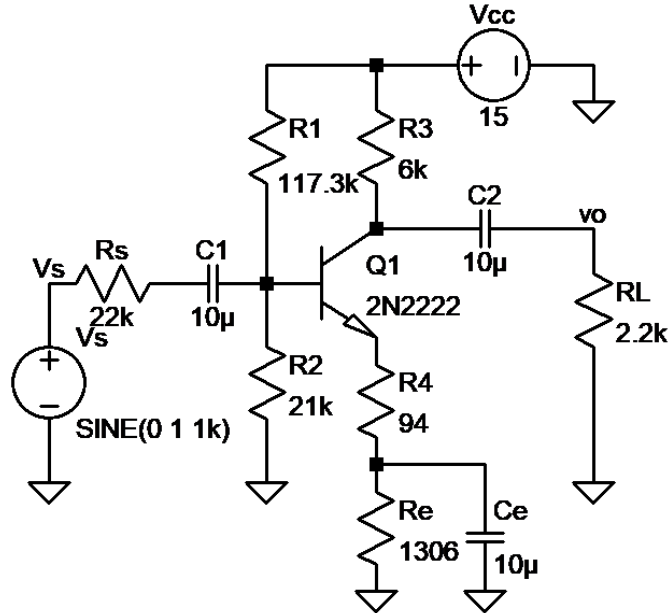
$$A_v = \frac{R_C}{r'_e + R_{e1}} = 50$$

Setting $V_{RC} = 6\text{ V}$ and $V_{RE} = 1.4\text{ V}$,

- $R_C = \frac{6\text{ V}}{1\text{ mA}} = 6\text{ k}\Omega$,
- $R_{e1} = \frac{R_C}{A_v} - r'_e = \frac{6000}{50} - 26 = 94\ \Omega$,
- $R_{e2} = \frac{1.4\text{ V}}{1\text{ mA}} - 94\ \Omega = 1306\ \Omega$.

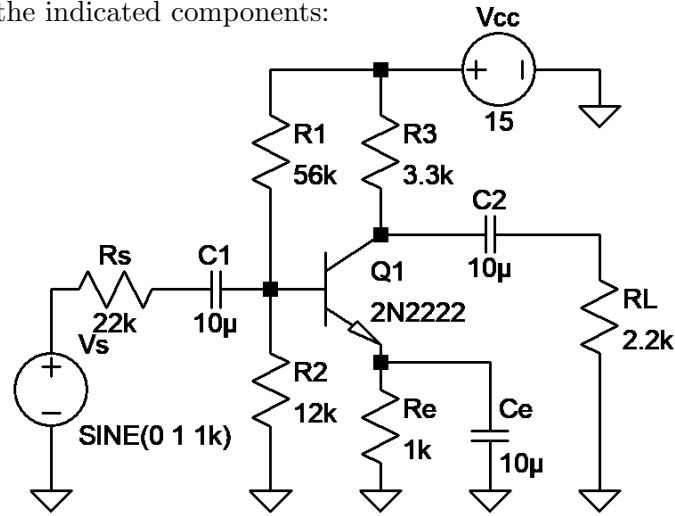
With $V_B = V_{RE} + V_{BE} = 2.1\text{ V}$ and $I_{div} = 10 \cdot I_B$,

$$R_2 = \frac{2.1\text{ V}}{100\ \mu\text{A}} = 21\text{ k}\Omega, \quad R_1 = \frac{15\text{ V} - 2.1\text{ V}}{110\ \mu\text{A}} = 117.3\text{ k}\Omega$$



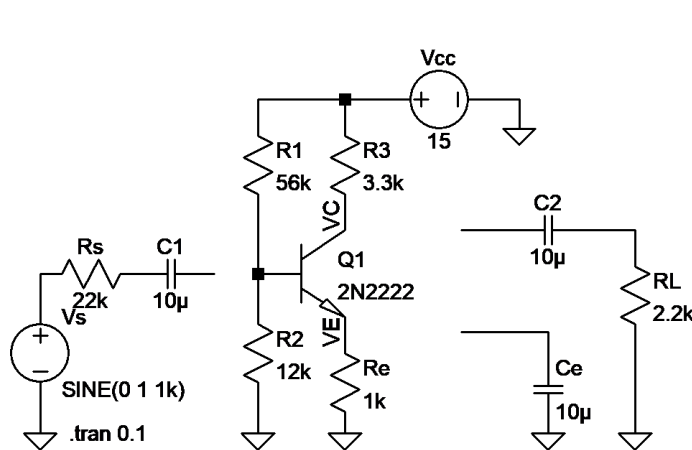
Simulation

1. The circuit with the indicated components:

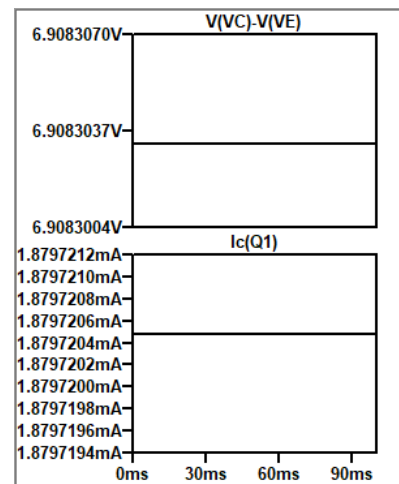


Circuit 1

2. Obtain the DC-equivalent circuit. $V_{CE,Q} \approx 6.91$ V and $I_{CE,Q} \approx 1.88$ mA.

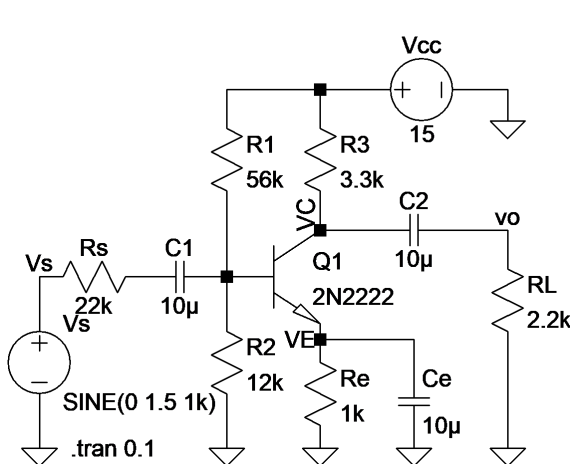


Circuit 2

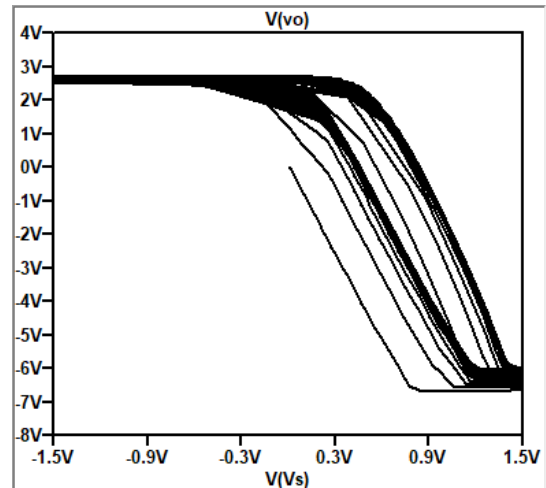


Graph 1 & 2

3. For $V_s = 0.5V_{pp}$, the amplitude becomes 1.5 V. The circuit and the graph:

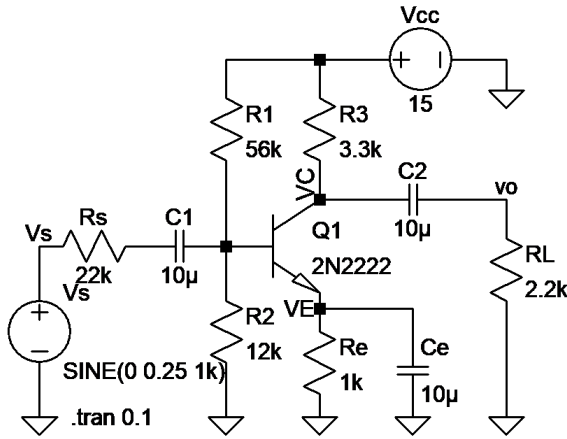


Circuit 3

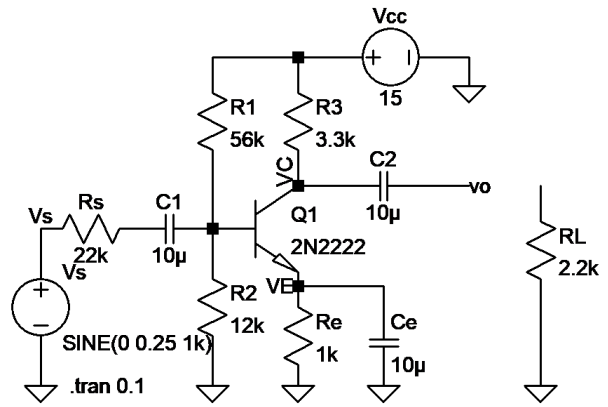


Graph 3

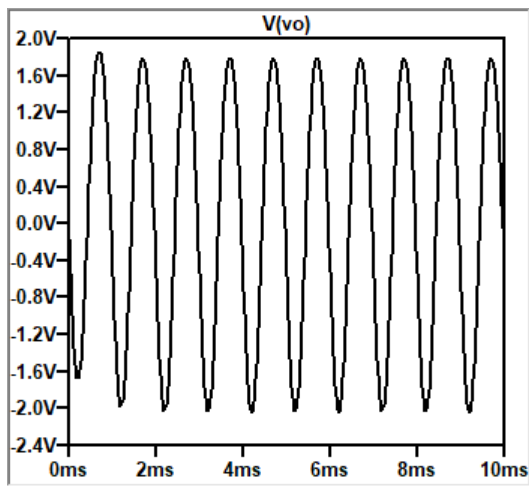
4. For $V_s = 0.5V_{pp}$, the amplitude becomes 0.25 V. The circuits and the graphs:



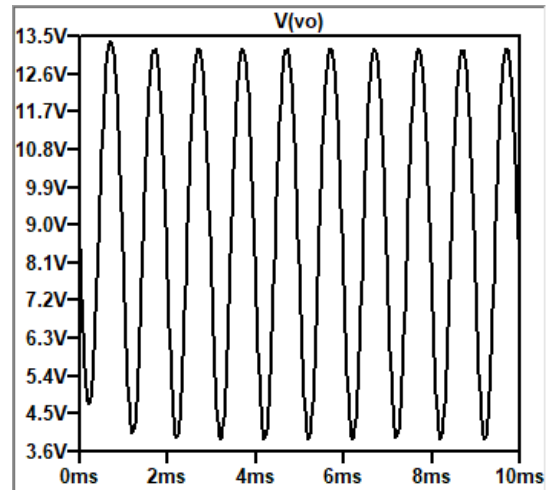
Circuit 4 with the load



Circuit 5 without the load



Graph 4: Circuit 4 graph



Graph 5: Circuit 5 graph